Code No: 121AF **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech I Year Examinations, March/April - 2023
COMPUTER PROGRAMMING
(Common to CE, EEE, ME, ECE, CSE, MIE)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a)	Define operator Precedence and Associativity.	[2]
b)	List out various data types used in c.	[3]
c)	What is a user defined function? Give example.	[2]
d)	How to initialize the elements of an array?	[3]
e)	Define Void Pointer?	[2]
f)	Write the syntax of calloc function.	[3]
g)	Write any two differences between text and binary file.	[2]
h)	Define self-referential structure?	[3]
i)	What is sorting?	[2]
j)	Give the importance of sorting.	[3]

PART - B

(50 Marks)

2.a) Define variable. List the rules for declaring variable. Give valid and invalid examples.
b) Write a program to find the Pascal's triangle for the given input. [5+5]

OR

3.a) Write a C program to display Fibonacci series using while loop.
b) Describe Switch statement. Write c program to read month number and display month name using switch case. [5+5]

4. How to pass parameters to functions? Explain each with an example. [10]

OR

5. Differentiate iterative functions and recursive function and write the recursive function to find the solution to towers of Hanoi problem. [10]

6. With suitable examples, explain different dynamic memory management functions in C. [10]

OR

7.a) How can you declare pointer variable? Differentiate a variable address and variable value. How to access variable value using pointers? Explain.
b) Write c program to reverse the given string without using strrev() function. [5+5]

8.a) Write a program for addition of two complex numbers by passing structures to a function.

b) Explain about the bit field in C. [5+5]

OR

9.a) Write a 'C' program to read the content of source file and convert all upper case letters in source file into lower case letters.

b) What are the file status functions? Explain their usage with a sample 'C' program. [5+5]

10.a) Write program to implement bubble sort.

b) Explain the procedure to insert a node into single linked list. [5+5]

OR

11.a) Describe the implementation of binary search.

b) Write in detailed note on queue data structure. [5+5]

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